

HOSPITAL MANAGEMENT SYSTEM

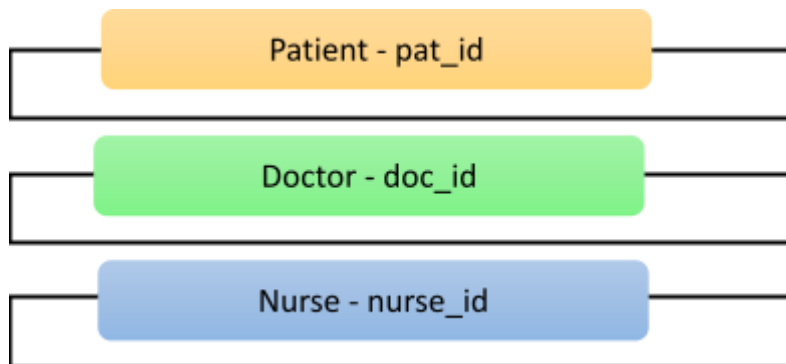
Abstract:

Hospital management system is designed to make the changes from manual system to automate with computerized technology which makes the data to be secured and required data can be taken out any time. This design helps the user to avoid record maintenance manually. We can store data for a longer period of time with easy access and manipulation of the data. Only the user and admin can access the data only with their credentials. This project describes how the data are stored, updated, deleted, etc. It deals with collection of patient information, diagnosis details, etc. The main function of the system is to register and store the patient, doctor, nurse and other details that can retrieve these details when required and also manipulate these data. System input contains patient details, diagnosis details where the system output is to get these details on the screen. This data is protected for personal use and makes the data processing very fast.

Introduction:

The hospital Management system project is all about storing the data about the registered patient in the system and also computerizing the billing in pharmacy and lab reports. This SQL oracle database gives a unique id for every registered patient and store each and every information about the patient, doctor and staff automatically. It includes a search facility to know the current status of each room. User can search availability of a doctor and the details of a patient using the id. The Hospital Management System, it is accessible either by an administrator or receptionist. Only they can add data into the database. The data can be retrieved easily. The data that are stored in a database are protected for personal use and it makes the data processing very fast. It is very powerful, flexible, and easy to use and is designed and developed to deliver real conceivable benefits to hospitals. Hospital Management System is designed for multispecialty hospitals, to cover a wide range of hospital administration and management processes. It is an integrated end-to-end Hospital Management System that provides relevant information across the hospital to support effective decision making for patient care, hospital administration and critical financial accounting, in a seamless flow. Hospital Management System is a software product suite which not only stores only the registered patient but also the data that is related to patient like the attend doctor, nurse, diagnosis reports etc. Those are required in hospital these are highly maintained and protected.

Table name and the primary keys in the table:



ER Diagram:

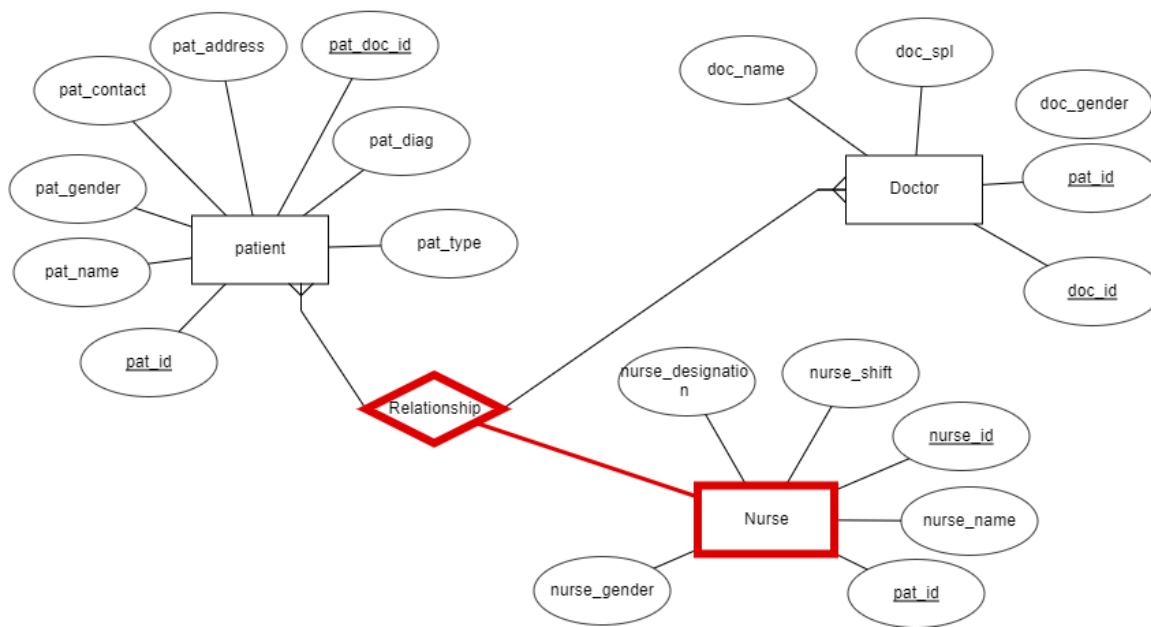


Table Structure:

s.no	Table name	Column name	Data type of the column	Size specified of the column	Constraint given	Relations it has
1	doctor	DOC_NAME DOC_GENDER DOC_SPL DOC_ID	VARCHAR VARCHAR VARCHAR VARCHAR	(25) (15) (25) (15)	<u>Primarykey</u>	Patient(pat_id)
2	nurse	NURSE_NAME NURSE_GENDER NURSE_DESIGNATION NURSE_SHIFT NURSE_ID	VARCHAR VARCHAR VARCHAR VARCHAR VARCHAR	(25) (25) (25) (25) (25)	<u>primarykey</u>	Patient(pat_id)
3	patient	PAT_ID PAT_NAME	VARCHAR VARCHAR	(15) (15)	<u>PrimaryKey</u>	

		PAT_GENDER PAT_CONTACT PAT_ADDRESS PAT_DOC_ID PAT_DIAG PAT_TYPE	VARCHAR VARCHAR VARCHAR VARCHAR VARCHAR VARCHAR	(15) (25) (100) (25) (15) (25)		
--	--	--	--	---	--	--

Patient Table:

Creation and Insertion in Patient Table:

Table for patient:

```
SQL> create table patient(pat_id varchar(15) primary key,pat_name varchar(15),pat_gender varchar(15),pat_contact varchar(25),pat_address varchar(100),pat_doc_id varchar(25),pat_diag varchar(15),pat_type varchar(25));

Table created.
```

Description of patient table:

```
SQL> desc patient;
Name                                Null?    Type
-----
PAT_ID                             NOT NULL VARCHAR2(15)
PAT_NAME                           VARCHAR2(15)
PAT_GENDER                         VARCHAR2(15)
PAT_CONTACT                        VARCHAR2(25)
PAT_ADDRESS                        VARCHAR2(100)
PAT_DOC_ID                        VARCHAR2(25)
PAT_DIAG                          VARCHAR2(15)
PAT_TYPE                          VARCHAR2(25)
```

Insert values in patient table:

```
SQL> insert into patient values('P001','Elijah','male',9895967845,'No:2,kk street,LM city','D003','Magraine','out patient');

1 row created.

SQL> insert into patient values('P002','Noah','male',9444558455,'No:3/4,RR street,VKC city','D001','Heartfailure','in patient');

1 row created.

SQL> insert into patient values('P003','Liam','male',9895945888,'No:47,QW street,CD city','D005','Asthma','out patient');

1 row created.

SQL> insert into patient values('P004','Olivia','female',9885967845,'No:2,SS street,UK city','D002','Plumonaryedema','in patient');

1 row created.

SQL> insert into patient values('P005','Emma','female',9895777845,'No:2,kk street,LM city','D004','Allergicrhinitis','out patient');

1 row created.
```

Doctor Table:

Creation and Insertion in Doctor Table:

Doctor table and insert values

```
SQL> create table doctor(doc_name varchar(25),doc_gender varchar(15),doc_spl varchar(25),doc_id varchar(15)primary key,pat_id varchar(15)references patient(pat_id));  
Table created.
```

Description of doctor table:

```
SQL> desc doctor;  
Name                               Null?   Type  
-----  
DOC_NAME                           VARCHA2(25)  
DOC_GENDER                         VARCHA2(15)  
DOC_SPL                            VARCHA2(25)  
DOC_ID                             NOT NULL VARCHA2(15)  
PAT_ID                             VARCHA2(15)
```

Insert values:

```
SQL> insert into doctor values('Dr.Ravi Sharma','male','Cardiologist','D001','P002');
1 row created.

SQL> insert into doctor values('Dr.William','male','Plumologist','D002','P004');
1 row created.

SQL> insert into doctor values('Dr.sophia','female','Neurologist','D003','P001');
1 row created.

SQL> insert into doctor values('Dr.Benjamin','male','Allrgist','D004','P005');
1 row created.

SQL> insert into doctor values('Dr.Harper','female','Asthma specialist','D005','P003');
1 row created.
```

Nurse Table:

Creation and Insertion in Nurse Table:

Nurse table:

```
SQL> create table nurse(nurse_name varchar(15),nurse_gender varchar(25)not null,nurse_designation varchar(25),nurse_shift varchar(25),nurse_id varchar(25)primary key,pa
t_id varchar(25)references patient(pat_id));
Table created.
```

Description of nurse table:

```
SQL> desc nurse;
Name                               Null?      Type
-----
NURSE_NAME                         NOT NULL   VARCHAR2(15)
NURSE_GENDER                       NOT NULL   VARCHAR2(25)
NURSE_DESIGNATION                  NOT NULL   VARCHAR2(25)
NURSE_SHIFT                        NOT NULL   VARCHAR2(25)
NURSE_ID                           NOT NULL   VARCHAR2(25)
PAT_ID                             NOT NULL   VARCHAR2(25)
```

Insert table:

```
SQL> insert into nurse values('Helen','female','LVN','morning shift','N001','P002');  
1 row created.
```

```
SQL> insert into nurse values('Dionne','female','RN','night shift','N002','P001');  
1 row created.  
SQL> insert into nurse values('Arya','female','RN','evening shift','N003','P005');  
1 row created.  
SQL> insert into nurse values('Taylor','female','LVN','morning shift','N004','P004');  
1 row created.  
SQL> insert into nurse values('Violet','female','RN','night shift','N005','P003');  
1 row created.
```

Constraint Violation:

- Check constraint,
- Not null constraint,
- Datatype violation,
- Size violation.

Check CONSTRAINT VOILATION:

```
SQL> alter table patient add check (pat_gender='male');  
alter table patient add check (pat_gender='male')  
*  
ERROR at line 1:  
ORA-02293: cannot validate (SYSTEM.SYS_C003877) - check constraint violated
```

NOT NULL VIOLATION:

```
SQL> insert into patient values(null,'Elijah','male',9895967845,'No:2,kk street,LM city','D003','Magraine','out patient');
insert into patient values(null,'Elijah','male',9895967845,'No:2,kk street,LM city','D003','Magraine','out patient')
*
ERROR at line 1:
ORA-01400: cannot insert NULL into ("SYSTEM"."PATIENT"."PAT_ID")
```

DATATYPE VOILATION

```
SQL> insert into nurse values('Helen','female','LVN','morning shift',N001);
insert into nurse values('Helen','female','LVN','morning shift',N001)
*
ERROR at line 1:
ORA-00984: column not allowed here
```

SIZE VIOLATION:

```
SQL> insert into nurse values('Helen','female','LVN','morning shift','N001');
insert into nurse values('Helen','female','LVN','morning shift','N001')
*
ERROR at line 1:
ORA-12899: value too large for column "SYSTEM"."NURSE"."NURSE_NAME" (actual: 5,
maximum: 1)
```

Primary and Foreign key constraints:

UNIQUE CONSTRAINT:

```
SQL> insert into patient values('P005','Mike','male',9895777845,'No:23,kkL street,LM city','D004','Allergicrhinitis','out patient');
insert into patient values('P005','Mike','male',9895777845,'No:23,kkL street,LM city','D004','Allergicrhinitis','out patient')
*
ERROR at line 1:
ORA-00001: unique constraint (SYSTEM.SYS_C003873) violated
```

INTEGRITY CONSTRAINT:

```
SQL> insert into doctor values('dr.Ravi Sharma','male','Cardiologist','D001');
insert into doctor values('dr.Ravi Sharma','male','Cardiologist','D001')
*
ERROR at line 1:
ORA-02291: integrity constraint (SYSTEM.SYS_C003874) violated - parent key not
found
```


Unique constraint violation:

```
SQL> insert into doctor values('Dr.Harper','female','Asthma specialist','D005','P003');
insert into doctor values('Dr.Harper','female','Asthma specialist','D005','P003')
*
ERROR at line 1:
ORA-00001: unique constraint (SYSTEM.SYS_C003909) violated
```

Integrity constraint:

```
SQL> insert into doctor values('dr.Ravi Sharma','male','Cardiologist','D001');
insert into doctor values('dr.Ravi Sharma','male','Cardiologist','D001')
*
ERROR at line 1:
ORA-02291: integrity constraint (SYSTEM.SYS_C003874) violated - parent key not
found
```

DDL commands (Select, Delete, Update, Alter):

Select command:

SQL> select * from patient;

PAT_ID	PAT_NAME	PAT_GENDER	PAT_CONTACT
PAT_ADDRESS			
PAT_DOC_ID	PAT_DIAG	PAT_TYPE	
P001	Elijah	male	9895967845
No:2,kk street,LM city	Magraine	out patient	
D003			
P002	Noah	male	9444558455
No:3/4,RR street,VKC city	Heartfailure	in patient	
D001			
PAT_ID	PAT_NAME	PAT_GENDER	PAT_CONTACT
PAT_ADDRESS			
PAT_DOC_ID	PAT_DIAG	PAT_TYPE	
P003	Liam	male	9895945888
No:47,QW street,CD city	Asthma	out patient	
D005			
P004	Olivia	female	9885967845
No:2,SS street,UK city			
D002			
PAT_ID	PAT_NAME	PAT_GENDER	PAT_CONTACT
PAT_ADDRESS			
PAT_DOC_ID	PAT_DIAG	PAT_TYPE	
P005	Emma	female	9895777845
No:2,kk street,LM city	Allergicrhinitis	out patient	
D004			

SQL> select * from doctor;

DOC_NAME	DOC_GENDER	DOC_SPL
DOC_ID PAT_ID		
Dr.Ravi Sharma	male	Cardiologist
D001	P002	
Dr.William	male	Plumologist
D002	P004	
Dr.sophia	female	Neurologist
D003	P001	
DOC_NAME	DOC_GENDER	DOC_SPL
DOC_ID PAT_ID		
Dr.Benjamin	male	Allrgist
D004	P005	
Dr.Harper	female	Asthma specialist
D005	P003	

```
SQL> select * from nurse;
```

NURSE_NAME	NURSE_GENDER	NURSE_DESIGNATION
------------	--------------	-------------------

NURSE_SHIFT	NURSE_ID	PAT_ID
-------------	----------	--------

Helen morning shift	female N001	LVN P002
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Dionne night shift	female N002	RN P001
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Arya evening shift	female N003	RN P005
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NURSE_NAME	NURSE_GENDER	NURSE_DESIGNATION
------------	--------------	-------------------

NURSE_SHIFT	NURSE_ID	PAT_ID
-------------	----------	--------

Taylor morning shift	female N004	LVN P004
-------------------------	----------------	-------------

Violet night shift	female N005	RN P003
-----------------------	----------------	------------

Delete command:

```
SQL> delete from patient where pat_name='Emma';
```

```
1 row deleted.
```

After delete command:

PAT_ID	PAT_NAME	PAT_GENDER	PAT_CONTACT
PAT_ADDRESS			
PAT_DOC_ID	PAT_DIAG	PAT_TYPE	
P001	Elijah	male	9895967845
No:2,kk street,LM city			
D003	Magraine	out patient	
P002	Noah	male	9444558455
No:3/4,RR street,VKC city			
D001	Heartfailure	in patient	
PAT_ID	PAT_NAME	PAT_GENDER	PAT_CONTACT
PAT_ADDRESS			
PAT_DOC_ID	PAT_DIAG	PAT_TYPE	
P003	Liam	male	9895945888
No:47,QW street,CD city			
D005	Asthma	out patient	
P004	Olivia	female	9885967845
No:2,SS street,UK city			
PAT_ID	PAT_NAME	PAT_GENDER	PAT_CONTACT
PAT_ADDRESS			
PAT_DOC_ID	PAT_DIAG	PAT_TYPE	
D002	Plumonaryedema	in patient	

UPDATE COMMAND:

```
SQL> update patient set pat_name='Elijah.A' where pat_id='P001';
1 row updated.
```

After update command:

```
SQL> select * from patient;
```

PAT_ID	PAT_NAME	PAT_GENDER	PAT_CONTACT
P001	Elijah.A	male	9895967845
No:2,kk street,LM city			
D003		Magraine	out patient
P002	Noah	male	9444558455
No:3/4,RR street,VKC city			
D001		Heartfailure	in patient
P003	Liam	male	9895945888
No:47,QW street,CD city			
D005		Asthma	out patient
P004	Olivia	female	9885967845
No:2,SS street,UK city			
D002		Plumonaryedema	in patient

Alter a table:

```
SQL> alter table patient add nurse_id varchar(25);
Table altered.
```

After alter table:

```
SQL> desc patient;
Name                                         Null?    Type
-----
PAT_ID                                     NOT NULL VARCHAR2(15)
PAT_NAME                                   VARCHAR2(15)
PAT_GENDER                                 VARCHAR2(15)
PAT_CONTACT                                VARCHAR2(25)
PAT_ADDRESS                                VARCHAR2(100)
PAT_DOC_ID                                 VARCHAR2(25)
PAT_DIAG                                   VARCHAR2(15)
PAT_TYPE                                   VARCHAR2(25)
NURSE_ID                                   VARCHAR2(25)
```

Error code and its explanation:

ERROR 1:

```
SQL> insert into doctor values('dr.Ravi Sharma','male','Cardiologist','D001');
insert into doctor values('dr.Ravi Sharma','male','Cardiologist','D001')
*
ERROR at line 1:
ORA-02291: integrity constraint (SYSTEM.SYS_C003874) violated - parent key not found
```

Error code is ORA-02291, integrity constraint is violated without giving any parent key . This error was corrected after inserting the parent key (pat_id).

After inserting the parent key

```
SQL> insert into doctor values('Dr.Ravi Sharma','male','Cardiologist','D001','P002');
1 row created.
```

ERROR 2:

```
SQL> insert into doctor value('Dr.Harper','female','Asthma specialist','D005','P003');
insert into doctor value('Dr.Harper','female','Asthma specialist','D005','P003')
*
ERROR at line 1:
ORA-00928: missing SELECT keyword
```

Error code is ORA-00928 : missing select keyword , it shows while we inserting the values using insert command “insert into table_name values(variable data type);” in this command we gave ‘value’ instead of ‘ values’ so that we encountered such error.

After error correction:

```
SQL> insert into doctor values('Dr.Harper','female','Asthma specialist','D005','P003');  
1 row created.
```

Conclusion:

Hospital Management System has been designed to achieve maximum efficiency and reduce the time taken to handle the hospital activity. It is designed to replace an existing record system. It helps the admin to shorten the time to record patient, doctor and nurse details, it also easy to add, delete or any changes and can be viewed any time.